



AMAS Landing in Banks

An industrial view

Xutong Liu · University of Twente

**UNIVERSITY
OF TWENTE.**

About me: software engineering for reliable LLMs



Xutong Liu

Postdoctoral researcher, University of Twente

- 2026 –** Postdoc, University of Twente (BMS)
- 2025 – 2026** Postdoc, Software Engineering Research Group, TU Delft
- 2019 – 2024** PhD in Computer Science, Nanjing University
- 2015 – 2019** BSc in Software Engineering, Dalian University of Technology

RESEARCH

Software engineering: empirical studies of software quality and testing

Now, with ING: reliability of RAG-based LLM systems (confidence and stability)



The UT × ING partnership on AI in Finance

- **2022 Partnership signed**
five-year AI and data science collaboration, at FinanceCom 2022 hosted by UT
- **2023 Training and network**
Doctoral Training School, COST FinAI links
- **2024 Digital Finance**
MSCA doctoral network in digital finance
- **2025 ING–UT workshop**
data analytics, quantitative models, model risk and AI, at ING Amsterdam
- **2026 LLM reliability**
my role: reliable RAG systems for banking



2022: UT and ING sign a five-year AI in Finance partnership

FINANCE

Digital employees, AI bootcamps: America's oldest bank is spending billions on tech

PUBLISHED MON, FEB 9 2026•6:00 AM EST | UPDATED MON, FEB 9 2026•9:47 AM EST



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A bank that already has digital employees

134

digital employees at BNY, America's oldest bank, alongside ~48,100 staff (Feb 2026)

EACH ONE HAS

-  a login and an email address
-  a human manager
-  performance reviews
-  tasks such as payment remediation and code repair

"I've just done three-quarters of the work for you ... in 10 minutes instead of ... two weeks."

— Robin Vince, BNY CEO, describing what a digital employee tells its manager

A digital employee does the due diligence on a new business client. It misses that the owner is on a sanctions list.

Who should answer for it?

- A** the agent itself
- B** the vendor that built the agent
- C** a named person in the bank who owns the agent
- D** someone else

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AN OPEN QUESTION There is no single right answer. Today regulators hold the bank responsible, even when a vendor built the system. Who inside the bank is for each bank to decide. BNY's choice: every digital employee has a human manager (slide 23).

By the end of this lecture you can...

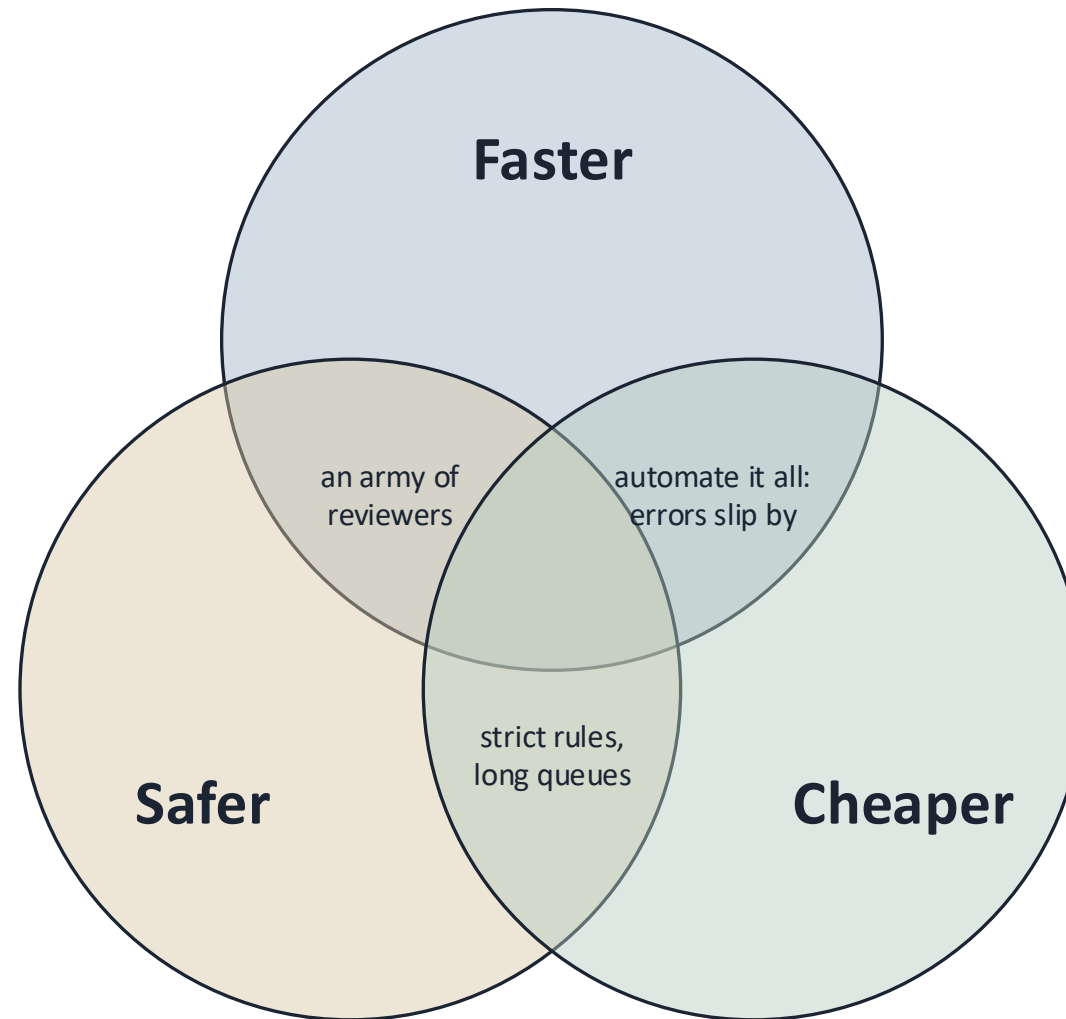
- 1 **Explain** why agent research in banking started early but is landing recently
- 2 **Describe** the practical problems bank scientists solved
- 3 **Analyse** how banks organise their agents today
- 4 **Explain** why the human in the loop is the bottleneck, and how confidence estimation helps

A bank is already a multi-agent organisation of people



Each has its own goal, sees only part of the picture, and hands work to the others.

Every bank workflow trades speed, safety and cost



Agents are entering banks, very fast

Aug 2025	● A global bank	KYC onboarding run by ten squads of AI agents (McKinsey report, bank not named)
Oct 2025	● BNY	Had over 100 "digital employees"
Early 2026	● NatWest	Started an agentic assistant that handles customers' fraud cases
Jun 2026	● FNBO with Nasdaq Verafin	Used agents that prepare financial-crime investigations

What changed: LLMs can read documents, use tools and write. That is the work banks run on.

As reported by banks, vendors and the press.

Software takes a bigger piece, the person moves up

WHAT SOFTWARE TAKES OVER, AND WHAT IS LEFT FOR THE PERSON

a rule

1988 · American Express

Person: judges every case,
helped by the rules

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a payment score

1992 · Falcon fraud detection

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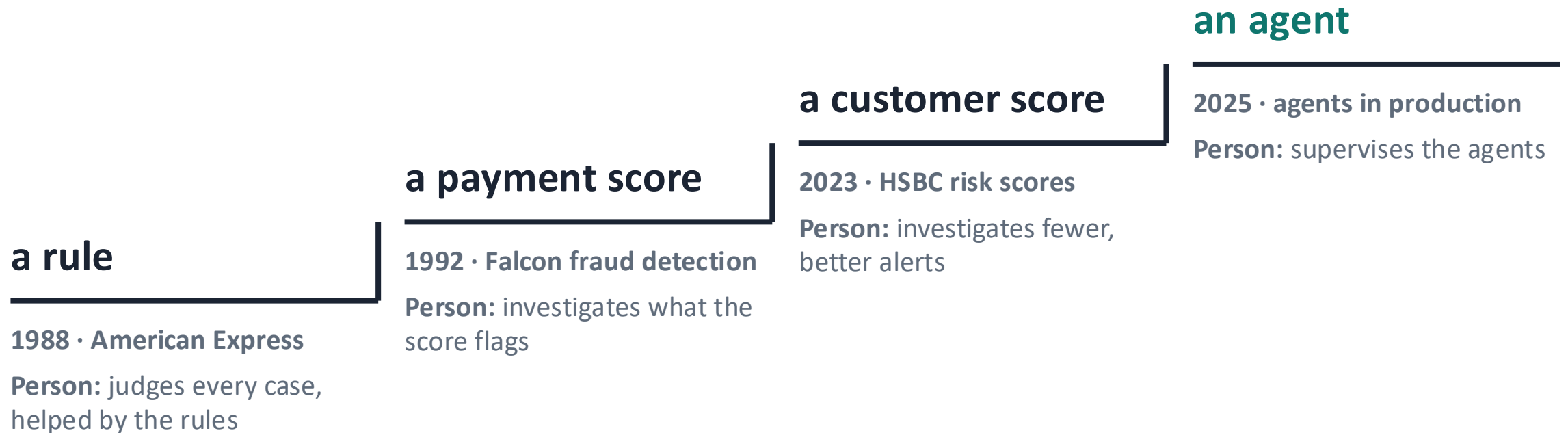
a customer score

2023 · HSBC risk scores

Person: investigates fewer,
better alerts

Software takes a bigger piece, the person moves up

WHAT SOFTWARE TAKES OVER, AND WHAT IS LEFT FOR THE PERSON



For most of this history, software took over steps inside a person's job, but rarely the job itself.

What LLMs change: software can now do the language work

Number work

amounts, dates, counts → a score
or an alert

Language work

reports, IDs, emails → evidence
and a case file

The decision

approve or reject, and own it

Until 2023

before LLMs

software

rules and scores



a person

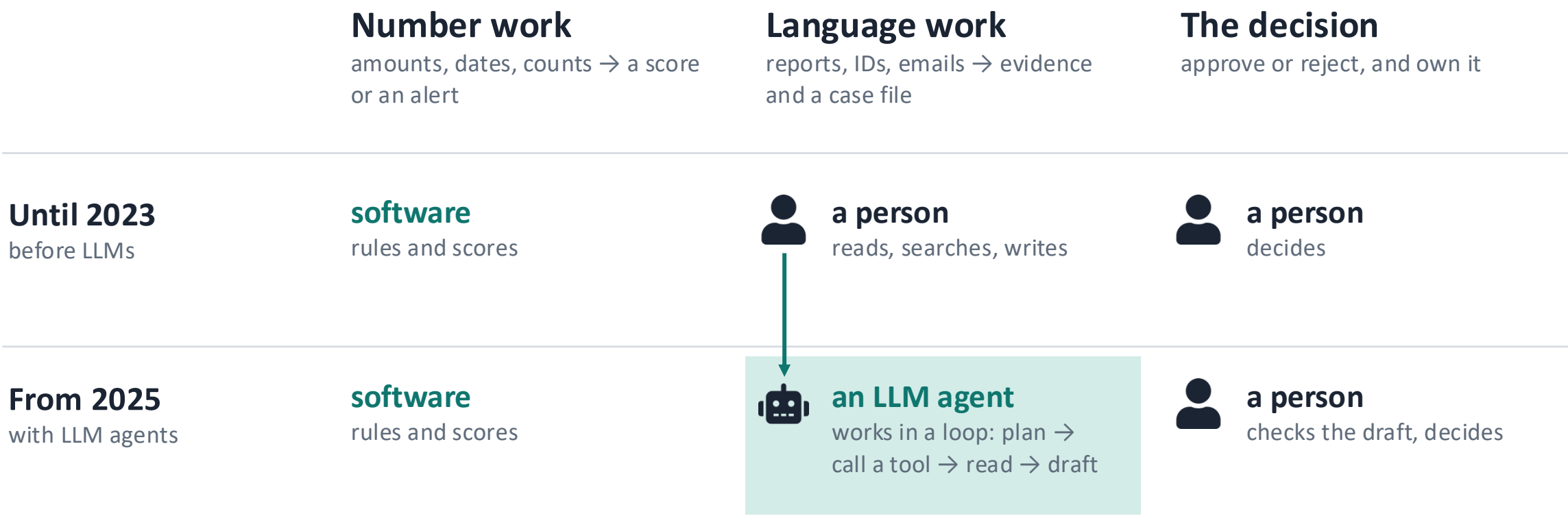
reads, searches, writes



a person

decides

What LLMs change: software can now do the language work

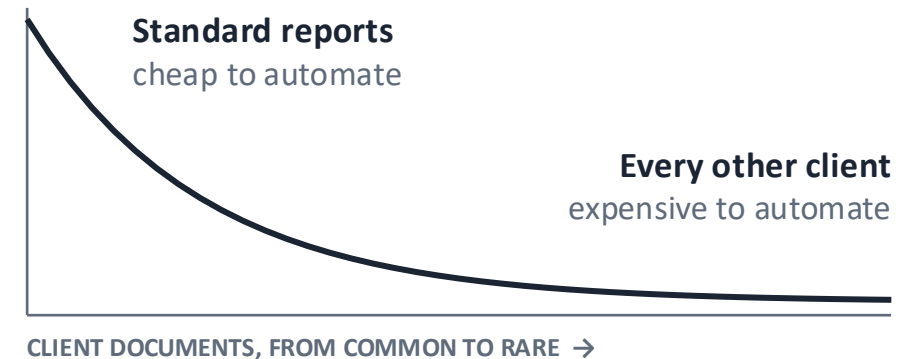


 **Only one cell changes: language work moves to LLM agents. The decision stays human.**

Why wasn't this automated before LLMs?



- OCR and NLP already existed, and worked for standard tasks.
- But a bank does not choose its input: **every client's annual report is different**. No shared format, no shared wording.
- So each new format meant new rules, new models, or manual work.



The problem was turning each client's different documents into a case-specific judgment.

So what was missing?

LLMs change the economics of automation

A general-purpose model that can understand, adapt, and write. Give it tools, and it becomes an AI agent.

BEFORE LLMS

One rule set per document type

Works on known formats

New format → new engineering

Pipeline fails

Analyst writes the rationale

WITH AN LLM

One model for every client's documents

Handles each client's wording and layout

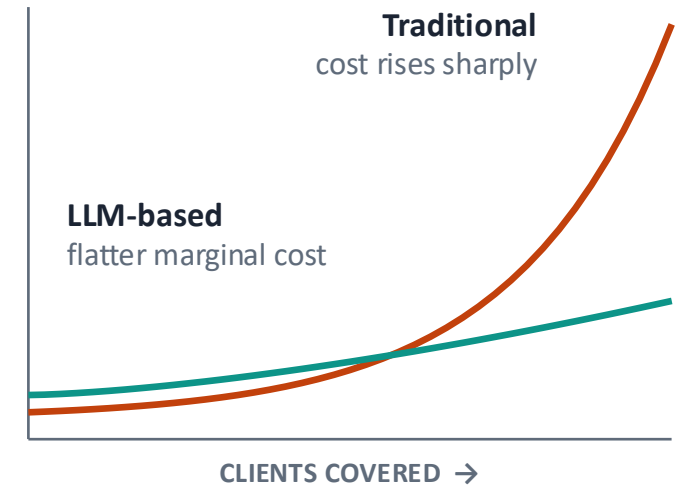
New format → the model adapts

WITH AN AI AGENT = LLM + TOOLS, IN A LOOP

Tries another source, tool or strategy

Agent drafts it, analyst decides

ENGINEERING COST



A sketch of the idea, not measured data.

The breakthrough is a **cheaper generalization** across the long tail of clients.

Why can one model read every client's report?



Scale

knows that "turnover", "net sales" and "revenue" mean the same thing

→ Attention

links a footnote on page 80 to a table on page 12

→ In-context learning

a new client format needs a new instruction, not a new model

→ Generation

writes the rationale that an analyst can check

One model for all clients.

2025–26: two ways banks put agents to work

A · a task

FNBO with Nasdaq Verafin



AGENTS COORDINATED BY
nothing: each works alone

THE PERSON
decides escalated cases

2025–26: two ways banks put agents to work

A • a task

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AGENTS COORDINATED BY
nothing: each works alone

THE PERSON
decides escalated cases

B • a process

a global bank



one KYC file, passed along

AGENTS COORDINATED BY
a fixed chain, squad to squad

THE PERSON
handles exceptions, 1 : 20+

A: agents that work alone, with a person behind each. B: agents that work together.

Pattern A: agents take single tasks for a human team

WORKFLOW 1 · SANCTIONS ALERTS, AS THEY COME IN



Sanctions agent

checks each alert against the list



closes clear false alarms itself, passes on the rest



Investigators decide

WORKFLOW 2 · SCHEDULED REVIEWS OF HIGH-RISK CUSTOMERS



EDD agent

prepares the review (enhanced due diligence)



recommends an outcome



Investigators review

Two separate workflows, one human team. The agents do not work together.

Why does each workflow get its own agent?

ONE DEPARTMENT, ONE AGENT

Customer due diligence



Its own agent

Sanctions



its own agent

Fraud



its own agent

Credit



its own agent

Nobody designed these agents from one AI architecture. Each one grew out of a department's own workflow.

The agent's boundary is the department's boundary.

Five things differ from one department to the next

- | | | |
|---|---------------------------|--|
| 1 | Goal | a different problem to solve, measured by different KPIs |
| 2 | Policy | its own rules, its own definition of risk, its own criteria to escalate |
| 3 | Data and output | different sources and fields; “risk score” does not mean the same thing |
| 4 | Trigger and timing | at onboarding, in real time, as a daily batch, or at a review every six months |
| 5 | Owner | its own users, budget, access rights, accountability and audit |

Many agents are not yet a multi-agent system.

Pattern B: squads of agents run the whole KYC process

TEN SQUADS, ONE STEP EACH. THE REPORT NAMES THESE EIGHT STEPS.



INSIDE EACH SQUAD: FOUR OR FIVE AGENTS

-  a lead agent
-  2–3 expert agents
-  a QA agent

 A human supervisor reviews the compiled file.

In this chain, every squad builds on the file from the squad before.

**Take the 8 steps. Say each is right 98% of the time.
How often is the whole file right?**

- A** about 98%
- B** about 92%
- C** about 85%

In this chain, every squad builds on the file from the squad before.

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ANSWER $0.98^8 \approx 0.85$: about 1 file in 7 has a mistake somewhere. Small errors add up along a chain. (The real bank uses ten squads: $0.98^{10} \approx 0.82$.)

A fixed chain of agents: what it gives, what it costs

STRENGTHS

Predictable

the order of the steps is fixed

Auditable

every step leaves a trail

Checked at every step

a QA agent in each squad

WEAKNESSES

Errors travel

a mistake in step 2 is still there in step 8

Rigid

a case that does not fit the steps needs a person

Hard to supervise

the person sees the file only at the end

The checks sit inside the chain. The person comes last.

The new rulebook leaves agents to the banks

2011 · SR 11-7, the US Federal Reserve's rules for bank models

every model must be validated, challenged and owned by a person: that is why the decision stayed human

2026 · SR 26-2 replaces it, and leaves agents out:

"Generative AI and agentic AI models are novel and rapidly evolving. As such, they are not within the scope of this guidance."

— US Federal Reserve, SR 26-2 (April 2026), which replaced SR 11-7

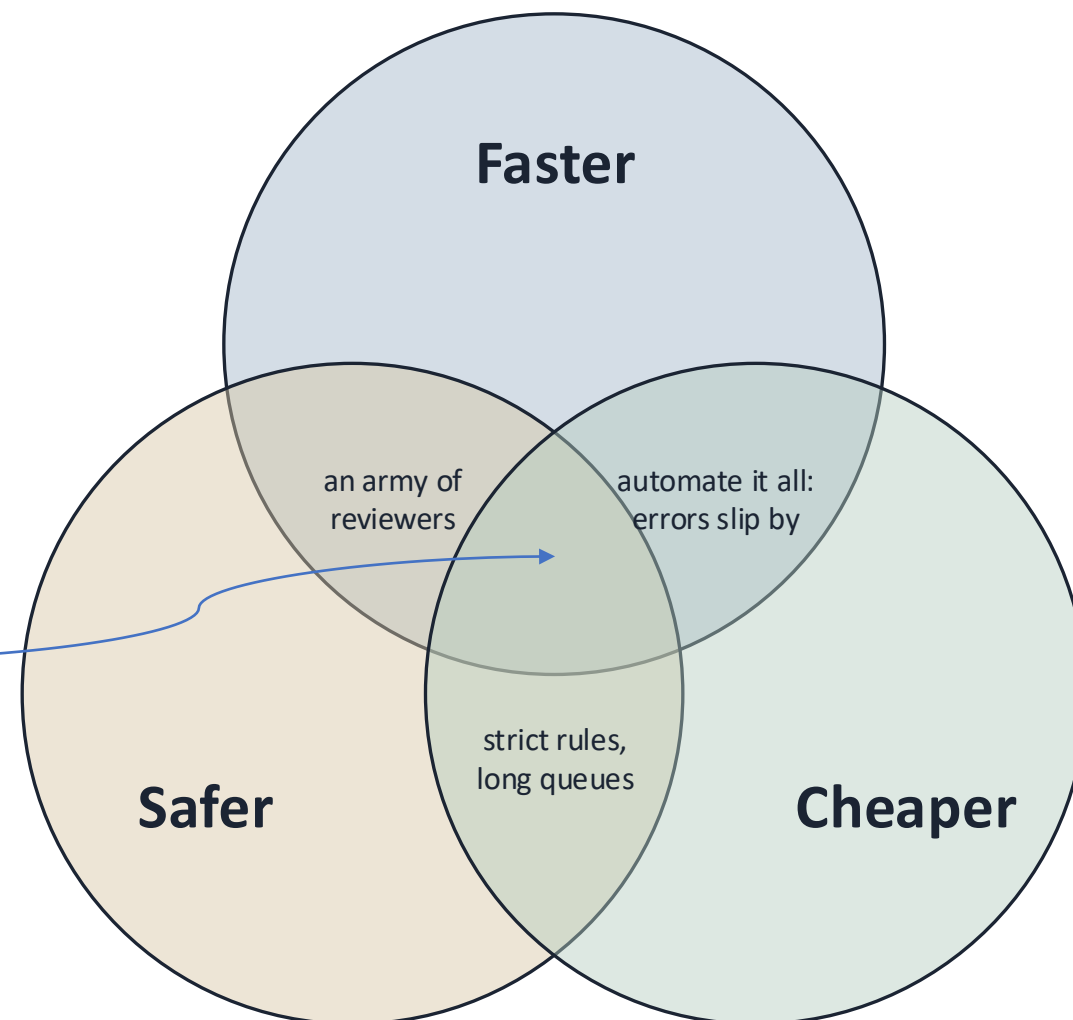
Banks must design their own controls for agents, starting with the human in the loop.

We need a confidence score that says which outputs to check first

TWO WAYS TO FAIL

- ❗ Check everything: the agents save no time
- ❗ Check nothing: errors pass straight through

BACK TO SLIDE 9 The middle of the triangle is a person who knows where to look.



How far can we trust one answer?

A RUNNING EXAMPLE

Question

What was the company's total debt at the end of 2020?

RAG answer

\$4.1 billion, citing two pages of the annual report

Confidence

a number for how likely this answer is right, e.g. 0.8

A GOOD SCORE DOES THREE THINGS

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3 Explains itself

a reviewer can see why the score is low, and where to look.

Four families of confidence estimation methods

ON OUR EXAMPLE

1 Logit-based

token probability, entropy (needs access to the model)

token probabilities of "\$4.1 billion" $\rightarrow$ 0.92

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verbalized confidence, $P(\text{True})$

"I am 95% sure" $\rightarrow$ 0.95

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"I am 95% sure" → 0.95

4 Evidence-based

entailment or an LLM judge over the retrieved sources

the cited page states \$4.1bn → supported

Each family has a trade-off

	Works with a closed API?	Extra cost	Shows why?	Main weakness
Logit-based	only if it exposes token probabilities	none	no	measures fluency, not truth

A fifth family trains a separate model to predict errors: often well calibrated, but it needs labelled data.

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Self-evaluation	yes	one extra prompt	it can give a reason, which may be made up	overconfident: scores cluster at 80–100%
Evidence-based	yes	one judge call per source	yes: points to the source	only as good as the retrieval and the judge

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CAST: does the evidence back the answer, or a rival?

1 Make a rival answer

change the key fact: "\$4.1 billion" → "\$3.2 billion"

Liu, Groot Beumer, Osterrieder & Baak (2026). CAST. FinNLP 2026. University of Twente & ING.

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2 Ask about each source

does this passage support the answer or the rival more?

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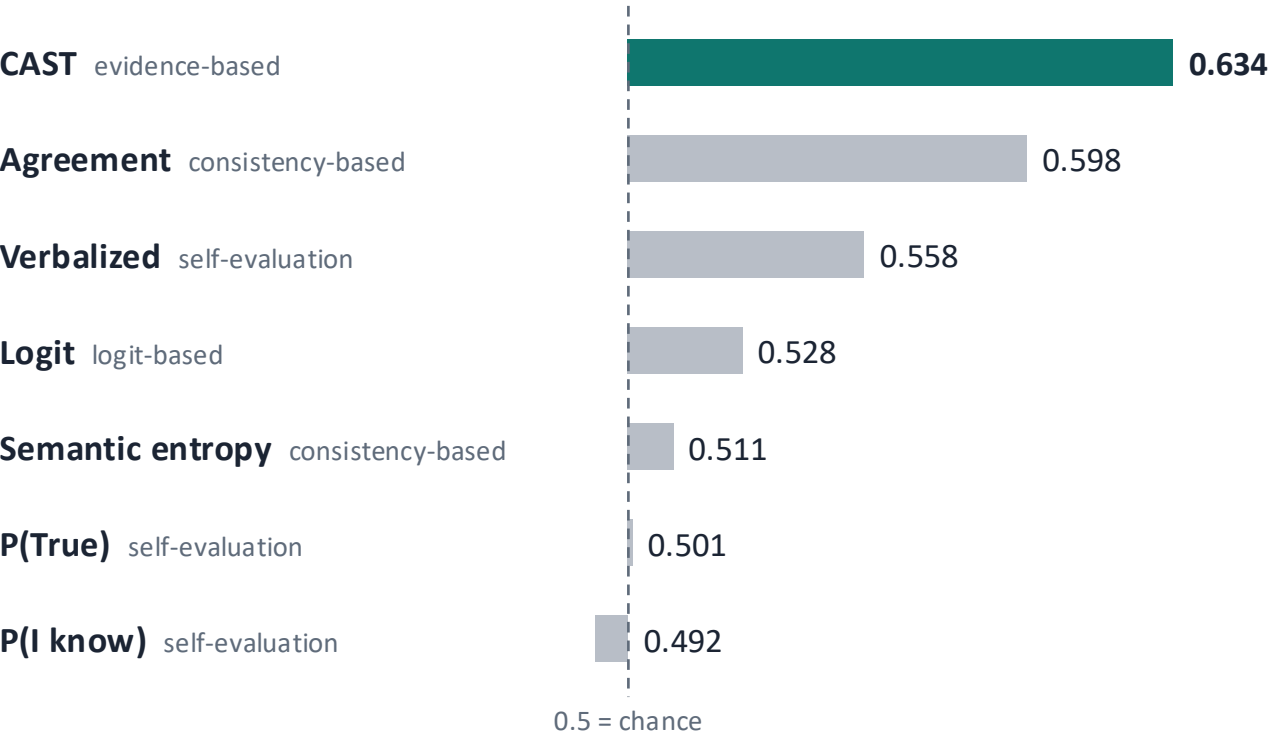
3 Score the gap

evidence favours the answer → high confidence; favours the rival → low, and those passages are shown first

Liu, Groot Beumer, Osterrieder & Baak (2026). CAST. FinNLP 2026. University of Twente & ING.

CAST ranks best, but there is room to grow

AUROC ON 1,286 QUESTIONS ABOUT FINANCIAL REPORTS



Bars start at chance. Liu et al., FinNLP 2026, Fig. 3 (MMDocRAG financial reports).

WHAT IT SHOWS

The evidence-based method ranks best here.
But every score is far from 1: knowing when an LLM is wrong is still hard.

Takeaways in confidence estimation

1 In banks, the answer comes from RAG

the model answers from the bank's own documents

2 A missed case is not acceptable

so we need to know when an answer may be wrong

3 There is still a lot to improve

the best method today is far from perfect, so people stay in the loop

Back to the learning objectives

THE FOUR GOALS FROM THE START, AS QUESTIONS

- 1** Why did agent research in banking start early, but land only recently?
- 2** Which practical problems did bank scientists solve, step by step?
- 3** How do banks organise their agents today?
- 4** Why is the human in the loop the bottleneck, and how does an evidence-based confidence score help?

Back to the learning objectives

THE FOUR GOALS FROM THE START, AS QUESTIONS

- 1** **Why did agent research in banking start early, but land only recently?**
→ Early agents could apply rules and compute scores, but could not read documents or write for people. LLMs can, and that language work is most of what banks do.
- 2** **Which practical problems did bank scientists solve, step by step?**
- 3** **How do banks organise their agents today?**
- 4** **Why is the human in the loop the bottleneck, and how does an evidence-based confidence score help?**

Back to the learning objectives

THE FOUR GOALS FROM THE START, AS QUESTIONS

- 1 Why did agent research in banking start early, but land only recently?**
→ Early agents could apply rules and compute scores, but could not read documents or write for people. LLMs can, and that language work is most of what banks do.
- 2 Which practical problems did bank scientists solve, step by step?**
→ First rules, then a score for each payment, then a score for each customer. Software took over judging payments and sorting alerts. The investigation stayed with people.
- 3 How do banks organise their agents today?**
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3 How do banks organise their agents today?

→ At least two ways: agents that work alone, each in its own department's workflow, with a person who decides. B: squads of agents in a fixed chain inside one process.

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→ One person looks after many agents and cannot check every output. A confidence score based on the evidence shows which answers to check first.



Two MSc thesis projects at ING

TOPIC 1

Stable evidence retrieval in RAG

The same KYC question, asked in different words, should retrieve the same evidence and reach the same conclusion. Today it often does not.

Measure and improve the stability of retrieval, evidence and answers.

Jointly supervised by UT and ING · hybrid, about 30 weeks · internship allowance · in Amsterdam one day a week

Looking for: strong Python, ML or statistics coursework, interest in LLMs and RAG

Interested? Email your CV to xutong.liu@utwente.nl

TOPIC 2

Evidence-based confidence for LLM answers

Reviewers need to know not only the answer, but how likely it is to be correct.

Use the retrieved evidence to estimate and calibrate confidence



Further reading

AGENTS IN BANKS TODAY

American Banker (2026). How FNBO uses agentic AI to investigate financial crime.

pattern A: the two agents at FNBO

McKinsey (2025). How agentic AI can change the way banks fight financial crime.

pattern B: the KYC squads

Axios (2025). This Wall Street bank has over 100 “digital employees”.

the BNY story

BEFORE LLMS

Gao, Xu, Wang & Green (2009). Knowledge-based anti-money laundering: a software agent bank application. *Journal of Knowledge Management* 13(2).

agents against money laundering, long before LLMS

HSBC (2023–24). Harnessing the power of AI to fight financial crime.

the step from a payment score to a customer score

WHO ANSWERS FOR THE AGENT?

US Federal Reserve, OCC & FDIC (2026). SR 26-2: Revised guidance on model risk management.

the guidance that leaves agents out

HOW FAR CAN WE TRUST ONE ANSWER?

Geng et al. (2024). A survey of confidence estimation and calibration in large language models. *NAACL*.

the method families, in depth

Liu, Groot Beumer, Osterrieder & Baak (2026). CAST: an interpretable evidence-based confidence estimation tool for RAG. *FinNLP*.

the method from this lecture

THE BIGGER PICTURE

ECB Banking Supervision (2026). Supervisory priorities 2026–28.

what supervisors expect banks to fix in the next years